Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves vertically to form an equation to find the tension.	AO3.1b	M1	$0.4 \times 9.8 = T\cos 30°$
	Finds correct tension.	AO1.1b	A1	$T = 4.5$ N
(b)	Finds radius.	AO1.1b	B1	$r = 0.6 \sin 30°$ $= 0.3$
	Forms an equation to find angular speed with 'their' radius and 'their' tension.	AO3.1b	M1	$T = \sin 30° = 0.4 \times r\omega^2$
	Obtains correct angular speed to 2 sf. FT 'their' values provided both M1 marks have been awarded	AO1.1b	A1F	$\omega = \sqrt{\dfrac{4.5 \sin 30°}{0.4 \times 0.3}}$ $= 4.3$ rad s^{-1}
(c)	States two appropriate assumptions.	AO3.5b	B1	Light and inextensible.
				Total 6 marks

Q2.

(a) For particle B,
tension in string = $2.1g$ N

B1

Resolve horizontally for particle A:

$m\omega^2 r = T$

Or $m_1\omega^2 r = m_2 g$ *or* $\dfrac{m_1 v^2}{r} = m_2 g$
(condone lack of 1 and 2)

M1

$1.4\omega^2 \times 0.3 = 2.1g$

$\omega^2 = 49$

A1

Angular velocity is 7 rad / sec

A1 **4**

(b) Using $v = r\omega$:
speed = 0.3 × 7

M1

= 2.1 m s^{-1}
Part (b) marks can be awarded in (a)

A1 **2**

(c) Time taken is 2 / ω
Or $\frac{2\pi r}{2.1}$

M1

= $\frac{2\pi}{7}$ = 0.898 sec
Accept $\frac{2\pi}{7}$
(0.895 M1A0)

A1 **2**

[8]

Q3.

(a) *Resolve vertically* $R = mg$
If the particle is on the point of sliding, $F = \mu R$
Ignore all inequalities

M1

$\therefore F = 0.3R = 0.3mg$

A1

Resolving radially: $F = m\omega^2 r$

M1

$0.3mg = m\omega^2 \times 0.8$

$\omega^2 = \frac{0.3 \times g}{0.8}$

$\omega = 1.92$

A1 4

(b) (i) 45 revolutions per minute = $\frac{90\pi}{60}$

M1

= $\frac{3\pi}{2}$ or 4.71 radians per second

A1 2

(ii) Resolving radially: $F = m\omega^2 r$

$m\,\mu\, g = m\left(\frac{3\pi}{2}\right)^2 \times 0.15$

M1A1 either side correct

A1

A1 second side correct

$\mu = \frac{\left(\frac{3\pi}{2}\right)^2 0.15}{g}$

$\mu = 0.340$

CAO (accept 0.339)

A1 4

[10]

Q4.

(a) 40 revolutions per minute
= 80π radians per minute

or $\frac{2}{3}$ *rev per second*

B1

= $\frac{4\pi}{3}$ radians per second

AG

B1 2

(b) Resolve vertically:
$T \cos 30 = 6g$
M1 1 term each side, 1 correct

M1A1

$T = 67.9$ N
AG

A1

3

(c) Resolve horizontally:
$T \sin 30 = m\omega^2 r$
M1 1 term each side, 1 correct

M1

A1 T sin 30

A1

$67.9 \sin 30 = 6 \times r \times \left(\frac{4\pi}{3}\right)^2$
A1 RHS

A1

$r = 0.322$ m
Condone 0.323 (using π as 3.14)

A1

4

[9]

Q5.

(a) (i) $T \cos 60° = 0.4\,g$

M1A1

$T = 7.84$ newtons
AG

A1

3

(b) $T \cos 30° = 0.4\,\omega^2 r$

M1A1

$r = \tan 60°$

B1

$7.84 \times \cos 30° = 0.4\ \omega^2 \times \tan 60°$

Subs

m1

$\omega = 3.13$

AG

A1

5

(c) $T = \frac{2\pi}{\omega} = 2.007$

≈ 2 sec

B1

1

[9]

Q6.

(a) Period = $\frac{2\pi}{\omega}$

M1

$\therefore\ 1.5 = \frac{2\pi}{\omega}$

$\omega = \frac{2\pi}{1.5} = \frac{4\pi}{3} = 1.3\pi$

Any – must leave π

A1

2

(b) (i) Acceleration = $r\omega^2$

$= 0.6\left(\frac{4\pi}{3}\right)^3$

Attempt to use formula

M1

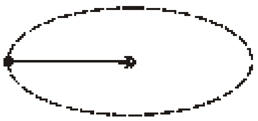
10.5 or $\frac{16\pi^2}{15}$

Follow through their ω

A1ft

2

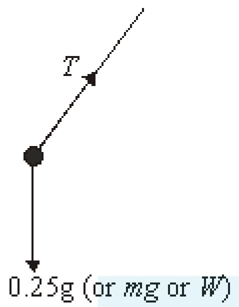
(ii)



B1

1

(c) (i)



(b)(ii) & (c)(i) could be on a single diagram

B1

1

(ii) Vertically (let $A\hat{B}O = \alpha$) $T \sin \alpha = mg$ (1)

Horizontally

Values may or may not be substituted in each equation throughout

M1A1

$T \cos \alpha = mr\omega^2$ (2)

ft $r\omega^2$ from b(i)

M1A1ft

(1) ÷ (2) $\tan \alpha = \dfrac{g}{r\omega^2}$

Dividing to get tan α or square and add to get T first

M1

$\alpha = \tan^{-1} \dfrac{9.8}{0.6\left(\dfrac{4\pi}{3}\right)^2}$ or $\tan^{-1}(0.9308)$

$\alpha = 43°$

Rounds to 43°

A1

6

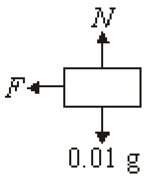
(d) No air resistance
Modelled as a particle

E1 1

[13]

Q7.

(a)



B1 1

(b) $F = mr\omega^2$

Formula quoted and attempt at use

M1

$= 0.01(0.4)\omega^2$
$= 0.004\omega^2$

A1 2

(c) $F = \mu N$
$= 0.8\ (0.01g)$

Use of limiting fraction

M1

$= 0.008g$

or 0.0784 seen

A1

Limiting, so
$0.004\omega^2 = 0.008g$

Equates their expressions

M1

$\omega^2 = 2g$
$\omega = 4.43$

AG

A1 4

[7]

Q8.

(a) Horizontally

$T \sin\theta = mr\omega^2$

Newton's Law horizontally

M1A1

$r = l\sin\theta$

$\Rightarrow$ T sin θ = ml sin $\theta\omega^2$

Use of r = lsinθ in equation

B1

$\Rightarrow T = ml\omega^2$

Cancelling to obtain answer

A1

4

(b) Vertically

$T\cos\theta = mg$

B1

$\Rightarrow (ml\omega^2)\cos\theta = mg$

Use of part (a)

m1

$\Rightarrow \cos\theta = \dfrac{g}{l\omega^2}$

A1

3

(c) (i) $T \leq 16$

$\Rightarrow (0.1)(0.4)\omega^2 \leq 16$

Use of part (a)

M1

$\omega \leq 20$

Withhold A1 if inequalities not evident

(use of ω_{max} or clearly explained conclusion accepted)

A1

2

(ii) Substitute in (b)

$\theta = \cos^{-1}\left(\frac{9.8}{0.4(20^2)}\right)$

Substitute in formula and use of $\cos^{-1}$

M1

= 86°

CAO

A1

2

[11]

Q9.

(a) 1100 r.p.m. = $1100 \times \frac{2\pi}{60}$ rad s^{-1}

Allow $\frac{110\pi}{3}$ *OE*

M1

≈ 115 rad s^{-1}

units not required

A1

2

(b) $115 = 8\dot{w}$

Allow $\frac{55\pi}{12}$ *OE*

M1

$\Rightarrow \dot{w} \approx 14.4$ rad s^{-2}

units not required

A1

2

[4]

Q10.

(a) 2π rads take 27.32 days

2π rads take 27.32 × 24 × 60 × 60 sec

days to seconds

B1

angular speed = $\frac{2\pi}{(27.32 \times 24 \times 60 \times 60)}$

radians per second

M1

≈ 2.66×10^{-6} rad s^{-1}

must see 4sf

A1

3

(b) Using $v = r\omega$:

= $(3.844 \times 10^8) \times (2.66 \times 10^{-6})$

use of formula

M1

≈ 1020 ms^{-1} (3sf)

AWRT

A1

2

[5]

Q11.

(a) (i) $R \cos 60° = 3 \times 9.8$

Resolving vertically

Correct equation

M1 A1

R = 58.8 N

Correct R

A1

3

(ii) $58.8 \cos 30° = 3 \times \frac{v^2}{0.5}$

Resolving vertically

Correct equation

M1 A1

$v = \sqrt{\frac{58.8 \cos 30°}{6}}$ = 2.91 ms^{-1}

Solving for v

Correct v

M1 A1

4

(b) (i) No change

No change

B1

1

(ii) Increased because v^2 is proportional to the radius

Increases

Reason

B1 B1

2

[10]

Examiner reports

Q2.

Many candidates resolved horizontally for particle *A* and obtained $m\omega^2 r = mg$ without appreciating that the m in $m\omega^2 r$ was 1.4, being the mass of particle *A*, whereas the m in mg was 2.1, being the mass of particle *B*. Other candidates found the speed, *v*, to be 2.1ms –1 but never found ω, or even included any equation involving ω.

Q3.

There were many good solutions to this question. A common problem seen in part (a) was candidates stating that $F = \mu R$ but being unable to notice that $R = mg$. Virtually all candidates could convert 45 revolutions per minute into an angular speed of $\frac{3\pi}{2}$ radians per second. Part (c) was usually completed well.

Q4.

In part (a), the conversion of 40 revolutions per minute into $\frac{4\pi}{3}$ radians per second was usually shown correctly. In part (c), candidates often forgot the m in the equation $T \sin 30 = m\omega^2 r$. A common error was in the use of $T \sin 30 = \frac{mv^2}{r}$, where candidates forgot that they were given the value of ω and not the value of r. Candidates were often unable to find correctly the numerical value of r when they had correctly obtained $r = \dfrac{67.9 \sin 30}{6 \times \left(\frac{4\pi}{3}\right)^2}$. Some gave the answer as 0.32 and it was surprisingly common for candidates to use π = 3.14 to find an answer of 0.323. Although this was accepted, it would have been far simpler for candidates to use the π button on their calculators.

Q6.

This question provided a better response than was expected. The structure seemed to help candidates to score well. A number of candidates did not seem to be aware of how to find the angular speed given angular displacement and time – part (a) therefore proved discriminating. The correct formula was often seen in part (b), although some substituted incorrectly. Diagrams were excellent and well labelled. Some candidates were well drilled in finding the angle, although some found the angle with the vertical because they had used the angle at the top. Follow through marks meant a critical error early on often only lost

Q7.

This question had a mixed response. In part (a), many candidates did not draw a correct force diagram, usually showing a fourth force or ignoring friction altogether. The structure of the question was there to help candidates, but some ignored the structure when answering parts (b) and (c). Full marks were duly awarded for this approach.

Q8.

Candidates who were familiar with the relevant book work performed well in this question. It was clear, though, that many candidates had not studied such work in sufficient depth. Many answers were faked to obtain the printed answer in part (b).

It was disappointing that some candidates could not substitute correct values in to the equation in part (b) to obtain the 2 marks in part (c) (ii).

Q9.

Most candidates were able to make good progress with this question.

Q10.

This was a good starter. Many candidates clearly knew the relevant formulae well. The accuracy policy meant that many candidates dropped a mark for failing to show a fourth figure in their final calculation of the angular speed.

Q11.

In part (a), almost all of the candidates produced the value of 58.8 N, but quite a few of these did not give a satisfactory argument to support their answer. Some candidates were clearly dividing the weight of the particle by the radius of the circle. In part (b), the common error was to simply use 58.8 without resolving to find the horizontal component. The answers to part (b) were very varied. Some candidates did produce reasonable answers without attempting part (a).